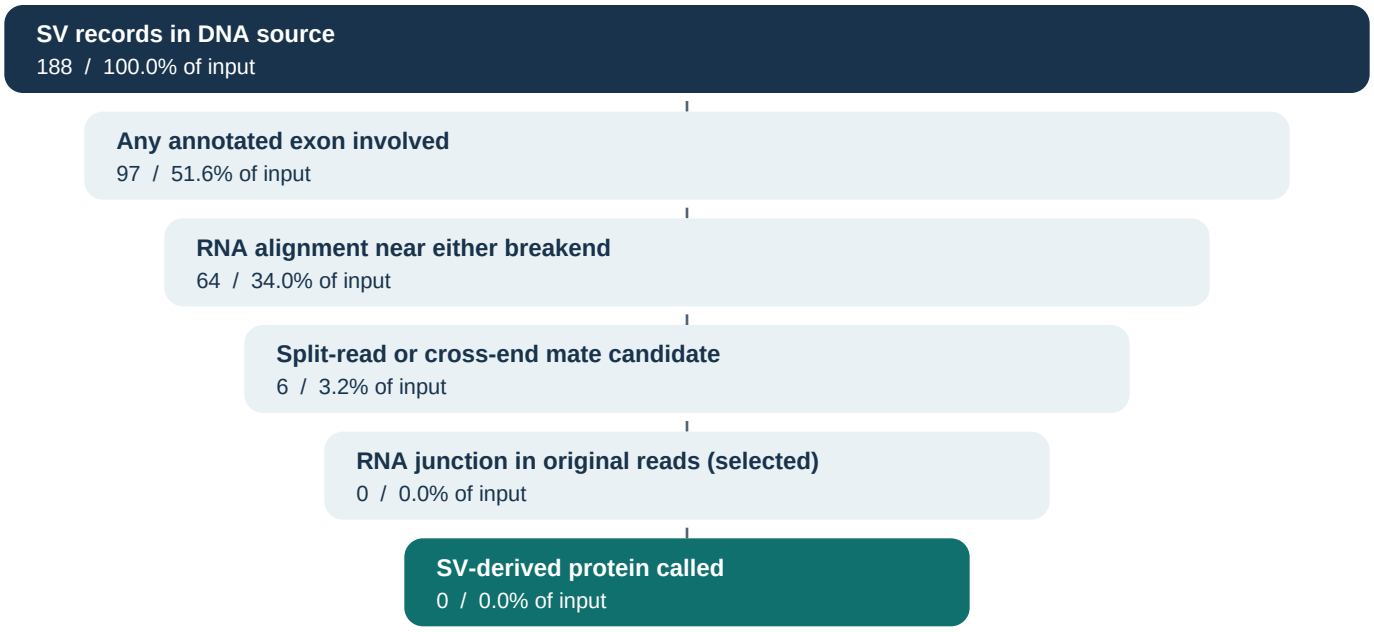
COLO829: somatic mutation and RNA evidence

Whole source-call audit | matched-normal DNA baseline | original tumor RNA | pinned reference ORFs. Counts below describe the specified public call sets and RNA run, not every biological mutation or a validated full-length isoform.

Small variants: DNA to RNA and local protein

Boxes show subset order; widths are schematic, not scaled to count.

Structural variants: DNA to RNA breakpoint evidence



Boxes show subset order; widths are schematic, not scaled to count.

Each funnel uses its own DNA source-record denominator. Breakend-window coverage is locus coverage, not evidence for the SV. A split-read or cross-end mate candidate links nominated windows but does not resolve junction bases, orientation, splice path or a translated product. The junction stage is a targeted original-read replay of selected events; it is a lower bound for the full source set.

At a glance

Evidence layer	Count	Interpretation
Ingested small variants	46,064	44,005 SNVs; 2,059 indels
Ingested SV source records	188	DNA breakpoint records/events, counted in their source sets
SVs involving an annotated exon	97	Span for DEL/DUP/INV; nominated breakpoint or insertion flanks otherwise
Any annotated exon involved	1,395	At least one changed base or insertion flank overlaps an Ensembl 87 exon
Ensembl 87 coding effect	373	Any annotated protein-coding transcript, reference model
orfdata ncORF coding-block overlap	5	Candidate reference ORF overlap; distinct from mutant translation
Original RNA ALT-observed calls	1,933	35,949 ALT reads; 35,617 ALT fragments
Calls with local protein contexts	124	12,218 ranked contexts; full-length isoforms unresolved

Coding annotation and its denominator

Of 46,064 DNA small-variant records, 373 change the coding sequence of at least one Ensembl 87 transcript. The pinned orfdata 0.34.1 GRCh38 catalog (7,264 reference ncORF models; release 2026-10-11.1) has exact coding-block overlaps at 5 calls. The groups overlap at 0 calls, so the union is 378. These are annotation opportunities. The catalog does not assess a cell-line-specific mutant ORF, RNA expression, or peptide evidence.

Reference annotation	All DNA calls	With original RNA ALT
Ensembl coding effect	373	180
orfdata coding-block overlap	5	2
Union of either category	378	see per-call table

SNVs query their changed reference base. Deletions query removed reference bases; insertions use the VCF anchor as a candidate overlap. The per-call table preserves that rule. A coding-block hit does not prove the inserted sequence is in frame.

Any annotated exon is involved in 1,395 calls (3.0% of the small-variant source set). This includes UTR and noncoding-transcript exons. Deletions query every removed reference base, including exon crossings with intronic endpoints; insertions query both neighboring reference bases as candidate exon involvement.

Original RNA and local protein evidence

RNA assay: SRR8615617, Illumina paired RNA. Each nominated small variant has a status from this run. An ALT read is a sequence observation; it is not a full-length mutant isoform.

RNA allele status	Small-variant records
alternate observed	1,933
reference only	1,637
reference and other only	36
other allele only	27
alignment without allele call	166
no alignment overlap	42,265

The 1,933 ALT-observed calls contribute 35,949 allele-read and 35,617 fragment observations. Isovar resolves at least one read-compatible **local** protein context for 124 calls (12,218 ranked rows). The remaining 1,809 ALT-observed calls have no local translation under the stated method. No-overlap and uncallable statuses cannot establish absence of RNA expression.

At CDKN2A chr9:21971154:CCG>C, all 6,984 ALT allele reads were evaluated. The top local frameshift context is **MMMGSARVAELLLLHGGAQLRRPR** with 5,338 supporting reads. Its 1,416 ranked local contexts reflect read-spanning alternatives, not full-length isoforms. TAGET polished representatives are an independent culture and assay; their cluster counts do not measure mutant allele support.

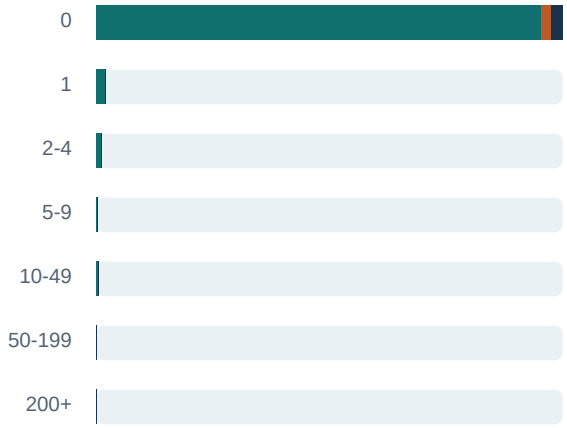
Structural-variant RNA screen

Breakend screen outcome	Source records
No breakend-window alignment	78
One end covered; no split candidate	36
Both ends covered; no split candidate	57
Cross-end mate context; path unresolved	11
Partner coordinate unresolved	6

The two SV benchmarks remain separate; no junction or SV-derived protein is called.

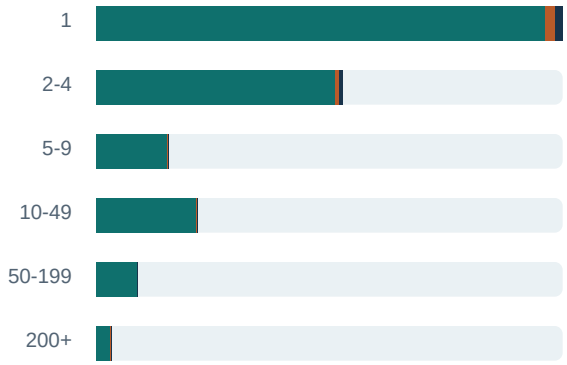
RNA alternate-read support by variant class

ALT reads per variant: all records



	SNV	insertion	deletion
0	42,140	922	1,069
1	917	19	16
2-4	487	10	7
5-9	144	4	1
10-49	204	3	2
50-199	83	2	1
200+	30	1	2

ALT reads per variant: positive support only

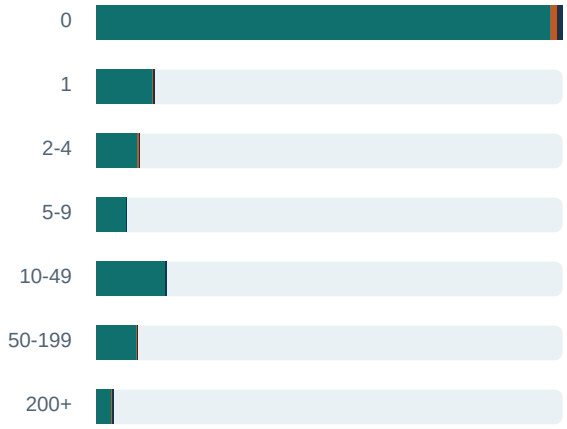


	SNV	insertion	deletion
1	917	19	16
2-4	487	10	7
5-9	144	4	1
10-49	204	3	2
50-199	83	2	1
200+	30	1	2

The positive-support panel excludes zero-ALT records. Bars stack SNV, insertion, deletion and other-replacement record counts. Matching colored columns print each class count directly. These are variant counts, not read counts.

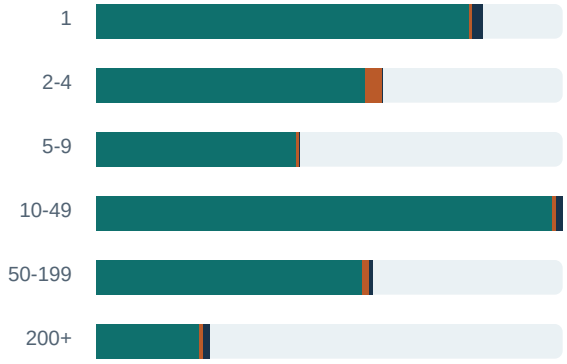
RNA ALT reads: exon-involved small variants

ALT reads per exon-involved variant: all records



	SNV	insertion	deletion
0	869	14	10
1	108	1	3
2-4	78	5	0
5-9	58	1	0
10-49	132	1	2
50-199	77	2	1
200+	30	1	2

ALT reads per exon-involved variant: positive support only



	SNV	insertion	deletion
1	108	1	3
2-4	78	5	0
5-9	58	1	0
10-49	132	1	2
50-199	77	2	1
200+	30	1	2

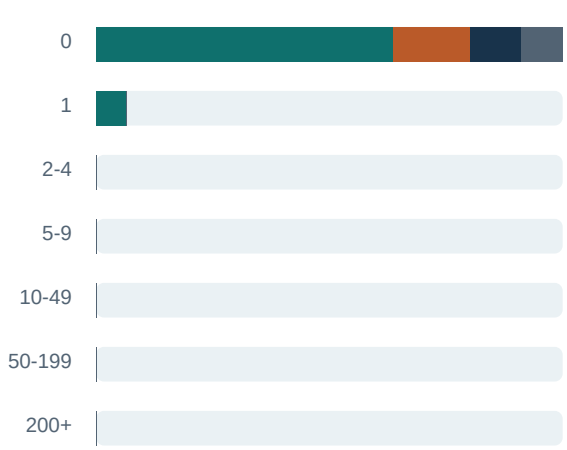
The input is the exon-involved subset of the small-variant source calls. The first panel retains zero-ALT calls; the second conditions on positive ALT reads. Insertion flanks are candidate exon involvement. SVs have a separate plot because a split alignment is not an ALT-spanning read.

Exon-involved nominated SVs: RNA breakpoint screen

97 of 188 nominated SV source records involve an Ensembl 87 exon. Intrachromosomal DEL/DUP/INV records query their full nominated span; INS queries insertion flanks; TRA/BND query nominated breakpoints. 4 records have insufficient breakpoint coordinates to rule out exon involvement and are excluded from this plot.

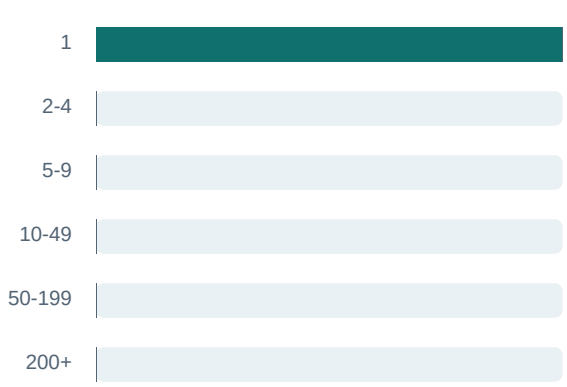
RNA screen	Exonic SV records	Candidate reads or pairs
Same-read split alignments	0	0
Cross-end primary mates	6	6

RNA link candidates per exon-involved SV: all records



	DEL	DUP	INV	BND
0	58	15	10	8
1	6	0	0	0
2-4	0	0	0	0
5-9	0	0	0	0
10-49	0	0	0	0
50-199	0	0	0	0
200+	0	0	0	0

RNA link candidates per exon-involved SV: positive support only



	DEL	DUP	INV	BND
1	6	0	0	0
2-4	0	0	0	0
5-9	0	0	0	0
10-49	0	0	0	0
50-199	0	0	0	0
200+	0	0	0	0

Bars count SV source records by reported class, binned on split-read plus cross-end mate candidate counts. Same-read supplementary alignments and opposite primary mates are different kinds of screening evidence; neither validates a junction sequence or translated product. The screen uses 500-base windows and mapping quality at least 20. The COLO829 benchmark call sets remain separate in the underlying table.

Original-read replay: junction versus RNA path

This selected-event replay uses oriented DNA adjacencies and actual RNA alignment records, including soft-clipped bases and low-MAPQ pieces. Direct junction fragments count reads crossing the same RNA join, even when their wider transcript sequences differ. MAPQ 20 is a sensitivity subset, not a gate on whether the junction sequence was observed. Isovar extends these joins through overlapping original read sequences; one read need not span the assembled context. Context length is a bounded assembly result, not an isoform-support count.

Nominated DNA record	Class	RNA join, all MAPQ	Join, MAPQ 20 subset	Longest context (bp)
SV3_DEL	DEL	3	0	1213
SV42_DEL	DEL	0	0	0
SV49_DEL	DEL	0	0	0

The chr1 SV3 deletion has six opposite-mate candidate pairs and three original fragments with an identical split junction sequence containing 19 unplaced bases. Its linked alignment pieces span MAPQ 1-52, including low-MAPQ supplementary pieces; the MAPQ 20 subset loses all three joins. Assembly extends this sequence to 1,213 bases through overlapping reads and reaches an extension limit. SV42 and SV49 have one mate candidate each but no directly observed join. SV3 is outside the exon-involved funnel above.

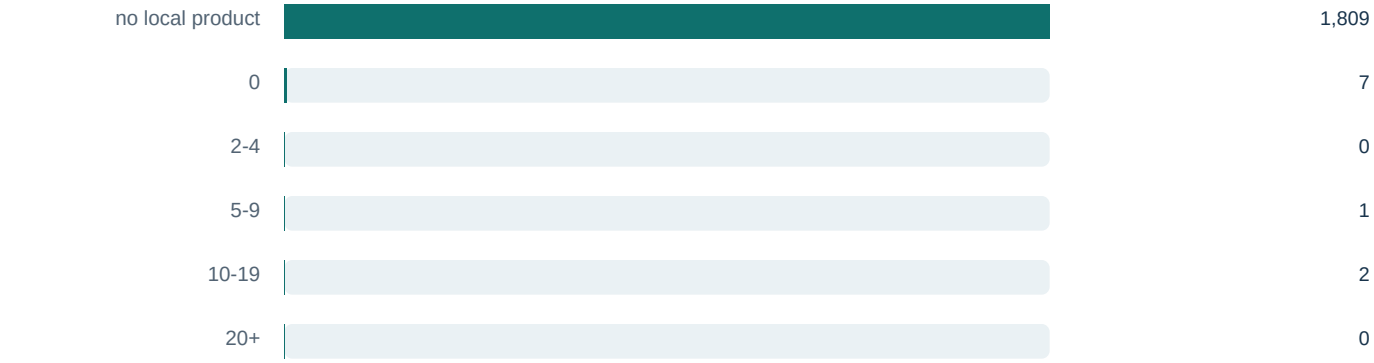
The HCC1395 replay BAM is a source-derived 309-record regional fixture; COLO829 uses the 129,534-record selected-site BAM. Both were extracted after aligning the complete original RNA run. No transcript models were supplied in this geometry audit, so frame and SV-derived protein status remain unresolved. Other splicing and DNA-to-RNA causation have not been excluded.

Isovar-marked amino-acid span

For each RNA ALT-observed call, count amino-acid positions in the mutation interval of its top-ranked local protein context. This is not an independently verified count of changed residues: synonymous DNA calls can produce a reference-identical amino acid window. Calls with no local context are shown explicitly. The second panel omits the one-residue category to expose the longer tail; its denominator is smaller. A zero-length interval can represent a deletion junction. These windows do not determine a complete frameshift tail or a full-length isoform.

Marked AA positions: all ALT-observed calls

Marked AA positions: excluding one-AA calls



Mutation classes and resolved outcomes

Category	DNA source	RNA ALT	Local context / product
DNA-predicted frameshift	5	4	See per-call local contexts
Reference ncORF coding-block overlap	5	2	Mutated ncORF not reconstructed
SV source record	188	Breakend screen only	0 resolved products

Frameshifts are Varcode top-effect predictions from DNA, not a count of RNA-confirmed full-length frameshift proteins. SV split/mate candidates do not count as productive until an RNA junction and protein path resolve.

Highest original-RNA ALT support with local contexts

Source call	Type	Support / self screen	Top local sequence
chr9:21971154:CCG>C	deletion	6,984 ALT reads	MMMGSARVAELLLLHGGAQLRRPR
chr2:105361307:G>GCCCC	insertion	847 ALT reads	FNCKKCSLSLVGRGFLTERDDILWGPRLWE...
chr6:3153911:G>A	SNV	694 ALT reads	MEFTEAESNMNDLVSEYQQYQDATAD

Least reference-self-like local contexts

Source call	Type	Support / self screen	Top local sequence
chr2:105361307:G>GCCCC	insertion	12/12 novel	FNCKKCSLSLVGRGFLTERDDILWGPRLWE...
chr14:32091979:C>T	SNV	9/9 novel	VYQNHVQHLISEKRRVEMKEKFKKILEKIQ...
chr2:197402055:G>A	SNV	9/9 novel	DEQQKVRTISALAIAALAEAATLYGIESFD...

The score is the fraction of distinct mutation-overlapping 9-mers absent from the pinned Ensembl 87 reference protein FASTA. It is an exact-match screen of these local windows, not patient-specific self, HLA binding, presentation or immunogenicity.

Highest supported non-SNV local products

Source call	Type	Support / self screen	Top local sequence
chr9:21971154:CCG>C	deletion	6,984 ALT reads	MMMGSARVAELLLLHGGAQLRRPR
chr2:105361307:G>GCCCC	insertion	847 ALT reads	FNCKKCSLSLVGRGFLTERDDILWGPRLWE...
chr14:92811641:GAGA>G	deletion	112 ALT reads	LAERKYSDEKRVDELQQQVKLYKLNLESSK...

The tables rank source records, not independent molecules. A displayed local context may represent only part of a translated protein and can be reference-identical for a synonymous allele.

Reference and mutant protein windows

The orange reference and teal mutant residues differ at the marked boundary; dark residues are shared flanks. Each reference window comes from the named Ensembl 87 transcript. The mutant is Isovar's top-ranked local context supported by tumor RNA reads. For frameshifts, downstream columns are anchored at the change, not a homology alignment or a complete protein. A dash means the reference protein ends before the mutant window.

chr9:21971154:CCG>C | deletion | 6,984 ALT reads | ENST00000304494 from amino acid 52

```
REF    MMMGSARVAELLLLHGAEPNCADP
MUT    MMMGSARVAELLLLHGGAQLRRPR
```

chr2:105361307:G>GCCCC | insertion | 847 ALT reads | ENST00000322142 from amino acid 249

```
REF    FNCKKCSLSLVGRGFLTERDDILCPDCGKDI----
MUT    FNCKKCSLSLVGRGFLTERDDILWGPRLWERHLNS
```

chr1:156265990:G>A | SNV | 586 ALT reads | ENST00000361813 from amino acid 525

```
REF    LEDMEEEEGTRSPTLEPPRGRSEAPD
MUT    LEDMEEEEGTRSPTLEPPRGRSEALD
```

Sources, processing and limits

Input	Pinned scope
Matched-normal small variants	SMaHT COLO829/COLO829BL, GRCh38 no-alt
SV benchmarks	Paulin Paz 2024 v2 (54 records) and Valle-Inclan 2022 lifted (134 records)
Original RNA	SRR8615617; 115,269,780 paired Illumina spots / 230,539,560 reads
Alignment	minimap2 2.26 -ax sr --frag=yes; NCBI GRCh38 no-alt

Reference model and interpretation

Small-variant coding effects use Varcode with Ensembl release 87 on GRCh38. The human orfdata 0.34.1 catalog uses Ensembl 101 GRCh38 reference ORFs (7,264 models, dataset 2026-10-11.1). An orfdata coding-block overlap is a reference annotation candidate; this analysis did not translate mutated ncORFs from the RNA reads. Genomic transcript overlap is not RNA-compatible isoform support. Original-read allele counts and local protein contexts are separate observations.

SHA-256 input pins

SMaHT SNV VCF gzip

`e57f77c8bc280bbad2255856c4062348f92f25da38726f367322c52736578f31`
SMaHT indel VCF gzip

`e8d0fb67925a2365b485cd65feab8dfdc12bbe54c0bc0bdec44c1846b518b685`
SV v2 GRCh38 VCF

`8ad075bf43aa71d89c591396569e813959ada44a30f171982c2d299411290bac`
Earlier SV lifted VCF

`a8f38454367cf8837a7dba1d29c9bc338a8e62cadfee6d2489931f8105f0dcdc`
SRR8615617 SRA archive

`9d89bb4f5a74fba7a8ba9495fd44046c35f4abb60d10c7bd9261f07569520b4f`
GRCh38 no-alt FASTA (uncompressed)

`9cce8b926416dd96b152deea85188495b75f7ac8d634cc723a017067be8702b7`
orfdata bundled GRCh38 catalog

`06bc0bf3f81c25a7356f71a38581ff5704c05dcb75752454bc1ca75075ab8c09`
Ensembl 87 GRCh38 protein FASTA gzip (self screen)

`d873c199d5dbf20869db3f2cbd6ac726259215d91c2782f9f1ddcad4bb117cf9`

Other antigen sources in this run

Category	Assessment
Viral RNA and viral coding paths	Not assessed: no viral-aligned RNA panel
Supplied fusion transcripts	Not assessed with direct junction reconstruction
Targeted antigen-gene isoforms	Not assessed with original-read isoform coverage
Mutated ncORF proteins	Not reconstructed; reference coding-block overlaps only

What remains unresolved

The SMaHT DNA, CCLE RNA and TAGET polished isoforms are separate cultures. The short-read genomic alignment is not splice-aware; junction-crossing reads may be lost. Three sequence-supported SV3 junction fragments have low-MAPQ linking pieces; placement, splice alternatives, full-length isoform linkage and SV-derived proteins remain unresolved.

Machine-readable per-call annotation, allele, protein, and SV tables and complete reconstruction commands are published with the Isovar HCC1395/COLO829 reports. These PDFs summarize selected public source call sets, not a complete catalogue of every cell-line mutation.